

Machine Learning Foundations and Deep Neural Architectures

Faculty of Computer Science and Artificial Intelligence

1. Gradient Descent Optimization (Inhidar Al-Tadarruj)

The parameter update equation for weight matrix W under gradient descent is:

$$W_{t+1} = W_t - \eta * \nabla_W L(W_t)$$

where η represents the learning rate and L denotes the scalar loss objective.

2. Mean Squared Error and Cross-Entropy Loss

$$\text{MSE Loss} = (1 / N) * \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$\text{Cross-Entropy Loss} = - \sum_{i=1}^C y_i * \log(\hat{y}_i)$$

Table 1: Activation function characteristics and vanishing gradient behavior.

Function	Equation	Range	Gradient at Extreme
----------	----------	-------	---------------------

Sigmoid	$1 / (1 + \exp(-z))$	$(0, 1)$	Zero (Vanishing)
---------	----------------------	----------	------------------

Tanh	$(\exp(z) - \exp(-z)) / (\exp(z) + \exp(-z))$	$(-1, 1)$	Zero (Vanishing)
------	---	-----------	------------------

ReLU	$\max(0, z)$	$[0, \infty)$	1 for $z > 0$ (Constant)
------	--------------	---------------	--------------------------

Figure 1 (Page 1): Multilayer perceptron architecture showing forward pass and backpropagation.